

A de Sitter–Unruh Modified-Inertia Field Theory, Complete Up To Its Constants: the Action, Its Radiative Protection, Its Equation Set, and Its Measurements

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Abstract

We assemble the de Sitter–Unruh modified-inertia (MI) program into a single classical action and report what that action does and does not deliver, at the standard of adversarially verified, exit-0 computation. The kernel at the center of the construction is not new: Milgrom (Phys. Lett. A 253:273, 1999) wrote the identical interpolation function from the same de Sitter-vacuum intuition, $\hat{\mu}(x) = \sqrt{1 + (2x)^{-2}} - (2x)^{-1}$, machine-verified identical to the kernel used here. What Milgrom left open, this paper supplies candidates for: the coefficient (his $\hat{a}_0 = 2cH_\Lambda$ versus this program's $a_0 = cH_\Lambda/Z$, $Z = \sqrt{32\pi/3}$, a factor $2Z \approx 11.58$) and the theory (he explicitly declined to endorse $\hat{\mu}$ for circular orbits absent a modified-inertia formulation). The action $S = S_{EH}[g] + S_u[g, u, \lambda] + S_{matter}[g, u, \psi; K] + S_{photon}[\tilde{g} = g + B[K]uu]$ carries 2 graviton + 2 photon + matter degrees of freedom and 0 frame degrees of freedom; it is ghost-free (including a genuine machine check of the nonlocal disformal sector), WEP-exact ($\eta = 0$), covariantly conserved on-shell, and $c_T = 1$ exact. The worldline-general identity $u_\mu \square_u u^\mu = -|a|^2$ makes the ring-exact RAR $g_{obs} = \sqrt{g_{bar}^2 + g_{bar}a_0}$ a derivation, not a fit. Two closure computations – the off-circular Wightman pullback (a null: the memory pole $\sqrt{H_\Lambda^2 + (a/c)^2}$ never descends below H_Λ) and a no-selection theorem (Gaussian KMS-passive baths are blind to the 4-point Jensen gap) – establish the paper's central honest claim: the theory is **complete up to five constants** $\{s, a_0, Z, \eta, \omega_c\}$, none of them derived. At one loop the interpolation is radiatively protected ($\delta\nu \equiv 0$ around dS; a_0 unrenormalized by an exact sum rule). At the solar system the theory's own ungated tail is excluded by 10^3 – $10^4\times$ and survives only through a gated crossover whose corner $\omega_c \in [1.78, 2.21] \times 10^{-14}$ rad/s (canonical footing) is an honest free fifth constant, two-sidedly falsifiable. On data: a gas-dominated SPARC estimator

built on the exact excess identity $g_{obs}^2 - g_{bar}^2 = a_0 g_{bar}$ returns $\hat{a}_0 = (0.92 - 1.18) \times 10^{-10} \text{ ms}^{-2} \pm 16\%$, from which the inversion $\Lambda = 3Z^2 \hat{a}_0^2 / c^4$ recovers the cosmological constant to a factor 1.1–1.6 of Planck; a concordance ledger shows the single Planck-fixed number threading every independent band with zero per-object freedom – while Λ CDM wins the raw information criteria on the same rotation curves, stated outright. One forced, η -independent, MG-impossible prediction survives everything: $d(\text{offset})/d(\text{radial anisotropy}) > 0$ for dispersion-supported systems. Every claim is backed by a committed, runnable script; the walls, the knife-edges, and the falsification exposures are listed with the results. **Erratum (v2): the disformal photon-metric route to dark-matter-free lensing is GW170817-excluded by ~6–7 orders – a genuine open failure of the lensing sector that leaves the dynamics sector untouched; see the Erratum note below.**

Erratum (v2, 2026-07-17)

After v1 (DOI 10.5281/zenodo.21403470) was posted, the photon-vs-graviton line-of-sight timing integral flagged in Section 2.4 as “order-of-magnitude satisfied only” was computed in full and adversarially verified (prep_2026/gw170817_check/, independent re-derivation both ways, exit-0). **The result is the opposite of that characterization.**

The disformal photon metric $\tilde{g} = g + B u_\mu u_\nu$ makes photons subluminal to gravitons by exactly $|c_\gamma - c_{GW}|/c = B/2$ (the graviton stays on g with $c_T = 1$; the $c_T = 1$ pass is therefore *irrelevant* to this bound, which constrains the graviton–photon *difference*). Accumulating B via $\nabla B = 4(\nu - 1)g_{bar}$ along the actual GW170817 sightline – source NGC 4993 at ~ 40 Mpc, with unavoidable host-galaxy and Milky-Way deep-MOND shell crossings – gives a differential arrival delay $\Delta t \sim 3.5 \times 10^7 \text{ s}$ (canonical) / $3.9 \times 10^7 \text{ s}$ (alternate), versus the observed 1.7 s bound; equivalently $|c_\gamma - c_{GW}|/c \sim 10^{-8}$ versus the GW170817 limit $< 10^{-15}$. **Excluded by ~6–7 orders of magnitude, robust across both a_0 footings and the entire sightline geometry** (best case a single thin host shell, ~ 5 orders; the void does not rescue it, because the delay depends on accumulated $B \sim 10^{-6}$, not on ∇B). B is tied to $(\nu - 1)g_{bar}$ by the lensing it must produce, so it cannot be shrunk: the framework’s disformal route falls in the same class as the bimetric/TeV-S lensing theories ruled out by GW170817 (Boran, Desai, Kahya & Woodard 2018).

Scope. This is a genuine open *failure of the lensing sector*, not a near-miss. It does **not** touch the dynamics sector: the RAR, BTFR, the single action’s constants, the graviton $c_T = 1$, the one-loop protection theorem, the equation set, the gas-dominated a_0 measurement, and the Λ -inversion are all independent of the disformal lensing construction and stand unchanged. Section 2.4 and the open-computations list (Section 9) are corrected accordingly. The honest standing of the lensing sector reverts to the program’s pre-paper position: relativistic,

dark-matter-free lensing is the framework’s deepest unsolved problem. A dark-matter-free lensing channel that clears GW170817 (single-metric nonlocal-MG, or a genuinely new construction) remains an open door, not a settled result.



1. Introduction: the wellhead, and what was missing

Any honest statement of this program begins with the credit. In 1999 Milgrom published a two-page paper (“The modified dynamics as a vacuum effect”, Phys. Lett. A 253:273, astro-ph/9805346) observing that a non-inertial observer in a de Sitter universe sees an Unruh temperature

$$T \propto \sqrt{a^2 + c^2 H_\Lambda^2},$$

and that the *excess* of that temperature over the inertial-observer floor, $\Delta T \propto \sqrt{a^2 + c^2 H_\Lambda^2} - cH_\Lambda$, defines an acceleration function with exactly the MOND limits. His equations (5) and (8)–(9) contain the interpolation

$$\hat{\mu}(x) = \sqrt{1 + (2x)^{-2}} - (2x)^{-1},$$

which is machine-verified identical (`equation_book/EQUATION_BOOK.md`, well-head credit; `sympy`) to the kernel this program uses, and which, inserted into the modified-inertia balance $a \mu(a/a_0) = g_N$, implies the exact excess identity $g_{obs}^2 - g_{bar}^2 = a_0 g_{bar}$ that Section 6 builds on. The parent law of everything in this paper is Milgrom’s, from the same de Sitter-vacuum intuition. The credit cuts both ways: the framework’s law carries Milgrom’s own dS-vacuum pedigree.

Milgrom’s 1999 note left two things missing, and this paper supplies a candidate for each:

1. **The coefficient.** Milgrom’s identification was $\hat{a}_0 = 2cH_\Lambda$. This program’s horizon-derived coefficient is $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3} = 5.78881$ – a factor $2Z \approx 11.58$ below Milgrom’s, and the value that the measurements of Section 7 confront. The numerical anchor, from Planck 2018 (TT,TE,EE+lowE+lensing): **canonical footing** ($\rho_{DE}, cH_\Lambda/Z$) $a_0 = (9.355 \pm 0.090) \times 10^{-11} \text{ m s}^{-2}$ (formal width 0.96%); **alternate footing** ($\rho_{total}, cH_0/Z$) $a_0 = (1.1305 \pm 0.0091) \times 10^{-10}$. The two footings differ by 21% and are not yet separated by any measurement; **both are carried on every dimensional number in this paper.** The value of Z itself is *postulated*, not derived – the $\kappa = 1/2$ closure was shown provably unforceable in earlier work, and nothing here changes that.

2. **The theory.** Milgrom explicitly declined to endorse $\hat{\mu}$ as the rotation-curve function for circular orbits, because no modified-inertia field formulation existed to license that step. Sections 2–5 of this paper present a candidate formulation: a single action in which the interpolation appears as a rigorously defined operator kernel, the circular-orbit reduction is a derivation from a worldline-general identity rather than an ansatz, and the boundary of what the action does *not* determine is computed rather than guessed.

What this paper is and is not. It is a results synthesis of the 2026-07-16 verification arc of the program, PhD-readable and step-by-step, with every load-bearing number traced to a committed, adversarially verified, exit-0 script (provenance table, Section 11). It is *not* a claim that the theory is finished, and it is not a unification claim of any kind: the program’s earlier over-claims in that direction were publicly retracted in June 2026 and are not reasserted. The strongest completeness language used anywhere below – and it is used deliberately, because it was earned by two closure computations – is: **the theory is complete up to its constants**. There are five of them, and none is derived.

Conventions: signature $(-+++)$; $y \equiv g_{bar}/a_0$; the framework’s own interpolation is always $\nu(y) = \sqrt{1+1/y}$, never the McGaugh RAR-fit ν (fitted $g^\dagger$ values are never compared across ν conventions); “canon”/“alt” label the two a_0 footings above.

2. The action

2.1 The single functional

One metric $g_{\mu\nu}$, one passive unit-timelike frame u^μ (the cosmic/dS-Unruh rest frame), one Lagrange multiplier λ , matter ψ , one kernel K , one scale a_0 :

$$S[g, u, \lambda, \psi] = S_{EH}[g] + S_u[g, u, \lambda] + S_{matter}[g, u, \psi; K] + S_{photon}[\tilde{g}]$$

with

$$S_{EH} = \frac{c^4}{16\pi G} \int \sqrt{-g} R \quad (\text{host gravity unmodified: 2 graviton dof})$$

$$S_u = - \int \sqrt{-g} \frac{\lambda}{2} (u^\mu u_\mu + 1) \quad (\text{passive frame, no kinetic term: 0 dof})$$

$$S_{matter} = - \frac{1}{2} \int \sqrt{-g} \rho_m [s u^\mu K(\square_u/a_0^2) u_\mu] \quad (\text{modified inertia, in the matter kinetic sector})$$

$$S_{\text{photon}} = -\frac{1}{4} \int \sqrt{-\tilde{g}} \tilde{g}^{\mu\alpha} \tilde{g}^{\nu\beta} F_{\mu\nu} F_{\alpha\beta}, \quad \tilde{g}_{\mu\nu} = g_{\mu\nu} + B u_\mu u_\nu \quad (\text{disformal lensing: 2 photon dof})$$

where

$$K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}}, \quad \square_u f = u^a \nabla_a (u^b \nabla_b f), \quad s = -1 \text{ (postulate)}.$$

The multiplier is fixed algebraically on-shell, $\lambda = -\rho_m s K$; the disformal coefficient B is fixed by the *same* kernel, $\nabla B = 4(\nu-1) g_{bar}$ with $\nu = 1/K$ – no new field, no new free function, no new propagating degree of freedom.

Kernel rigor (derived). K is a Herglotz–Nevanlinna function with a *unique* positive Borel measure: closed-form densities $\rho_A = (1 - \sqrt{1-4|t|})/(2\pi\sqrt{|t|})$ on $-1/4 < t < 0$ and $\rho_B = 1/(2\pi\sqrt{|t|})$ on $t < -1/4$, additive constant $a = 0.65411$, $\|K\| \leq 1$, causal-retarded. The sum rule

$$\int \frac{d\mu(t)}{|t|} = K(\infty) - K(0) = 1 \quad (\text{region-B share } 2/\pi \text{ exact; total } = 1 \text{ to } 10^{-8})$$

with $K(0) = 0$ means a_0 enters *only* through the argument $X = |a|^2/a_0^2$ and carries no counterterm at scale a_0 (Section 4).

2.2 The stress tensor, WEP, and conservation

Matter couples minimally to the single metric g (rods, clocks, and photons-in- g -sense ride g); modified inertia is a universal *inertial-scalar dressing* $W = s u^\mu K(\square_u/a_0^2) u_\mu$ of the matter kinetic term – matter feels the kernel through its own 4-acceleration via $X = |a|^2/a_0^2$. The three variations of the action give:

- $\delta/\delta\lambda$: the unit-timelike constraint $u \cdot u = -1$.
- $\delta/\delta u^\mu$: a frame equation whose algebraic source $J_\mu = -[\rho_m s K + \lambda] u_\mu$ is strictly $\ell = 0$ (transverse projection verified zero) and soaked by $\lambda = -\rho_m s K$ with no tertiary constraint tower. The derivative (K') terms are $\ell = 1$ worldline dynamics and never populate the $\ell = 2$ traceless shear – the structural evasion of the AeST Cassini Bianchi lock.
- $\delta/\delta g^{\mu\nu}$: the closed-form matter stress tensor

$$\boxed{T_{\mu\nu} = \alpha u_\mu u_\nu + \beta g_{\mu\nu} + \gamma a_\mu a_\nu}, \quad \alpha = -\rho_m s K, \quad \beta = \frac{1}{2} \rho_m s K, \quad \gamma = \frac{\rho_m s K'(X)}{a_0^2}.$$

In the principal (UV) limit $K \rightarrow 1$, $K' \rightarrow 0$: the anisotropic term vanishes and $T_{\mu\nu}$ is an isotropic perfect fluid – **no gravitational slip** ($\Psi = \Phi$), the AeST no-slip property recovered from modified *inertia* with no dynamical aether. The only slip-capable stress is $\gamma a_\mu a_\nu$, nonzero only in the MOND/IR regime; its exact IR magnitude is the open lensing-enhancement residual (it coincides with the closure gap of Section 3).

WEP is exact: $\eta = 0$, **derived.** W is built only from $(u, g, \partial u)$ and carries no matter-species label; every body’s inertial coefficient is the same function $\mu_{fw}(|a|/a_0) = K(X)$, and the balance inverts species-independently. Machine check: $\eta < 10^{-12}$ over $y \in [10^{-2}, 10^2]$ (`matter_coupling_Tmunu.py`, 14/14, exit 0). This is confronted with MICROSCOPE in Section 7.4.

Conservation, verified two ways. $\nabla_\mu T^{\mu\nu} = 0$ on-shell by diffeomorphism invariance: (i) the generic canonical Noether identity $\partial_\mu \Theta^\mu{}_\nu = -(EL) \partial_\nu \phi$ holds identically (symbolic, exact); (ii) instantiated with the actual MI acceleration kernel, the residual is 2.6×10^{-16} at 12 random spacetime points. The unit-norm constraint is dynamically preserved ($u \cdot a = 0$); the frame stays passive under the matter coupling.

2.3 Degrees of freedom and ghost-freedom of the full coupled system

The full coupled census (`wellposed.py`, 28/28 checks, exit 0, both footings):

property	verdict
propagating content	2 graviton + 2 photon + standard matter; frame = 0
Ostrogradsky ghost	none: local action first-order in every dynamical field; nonlocal K ghost-free by the Herglotz single healthy pole
Hamiltonian bounded below	yes: $\mu = K \in (0, 1]$; $\tilde{g}$ Lorentzian and $c_\gamma^2 = 1 - B > 0$ for $B < 1$ ($B \sim 6\text{--}7 \times 10^{-7}$ at a galaxy, both footings)
hyperbolicity	block-diagonal principal symbol, real characteristics; well-posed Cauchy problem (retarded history data)
causality	photon cone nested inside the g cone iff $B \geq 0$ iff $s = -1$; retarded kernel; single global causal order
c_T	$= 1$ exact: the graviton rides g , and $B u_\mu u_\nu$ has zero spatial ij block in the rest frame (genuinely computed, not asserted)

Three points deserve step-by-step emphasis.

The frame stays passive under the matter coupling (new result of this arc). The unit-norm second-class Dirac pair $\chi_1 = u \cdot u + 1$, $\chi_2 = u \cdot \pi$ has bracket $\{\chi_1, \chi_2\} = 2(u \cdot u)$, determinant $4(u \cdot u)^2 \rightarrow 4$ on-shell – and this is a *kinematic* bracket, unchanged when the momentum carries an arbitrary matter contribution. The frame principal symbol $(u \cdot k)^2 \rightarrow k_0^2$ is a transport ODE (double root at $k_0 = 0$ independent of spatial k , zero group velocity): the $K(a^2)$ dressing structurally cannot promote u to a propagating aether.

The modified-inertia Ostrogradsky trap is evaded, and the nonlocal disformal sector was checked for real. A naive worldline Lagrangian $L(\ddot{x})$ is Ostrogradsky-sick; here the acceleration is $a^\mu = u^\nu \nabla_\nu u^\mu$, a first-order gradient of the *field* $u(x)$, so the higher-derivative hypothesis is never met. Separately, an earlier machine check of the *nonlocal* disformal coefficient $B[K(\square_u)]$ was found to be tautological and was replaced by a genuine verification (`mi_closure_pin/ostro_nonlocal_verify.py`, 13/13, no hard-coded booleans): the exact Herglotz spectral representation gives an auxiliary tower with Ostrogradsky Hessian $\partial^2 L / \partial \ddot{\chi}^2 = 0$, kinetic Hessian $2d\mu(t) > 0$, masses $ta_0^2 > 0$ – healthy massive scalars, not a ghost – and the same machinery correctly *flags* a textbook $\ddot{q}^2$ ghost and a negative-measure kernel (live negative controls), so the check is discriminating, not decorative.

Lensing is forced into a separate photon coupling – the double-count obstruction. With one metric, putting the RAR enhancement into g and re-solving the same MI equation of motion forces $\nu_g = 1$ (solved symbolically): the enhancement cannot be shared between dynamics and light. A photon-sector coupling is therefore structurally mandatory; the disformal $\tilde{g} = g + Buu$ with B fixed by the same kernel is the *minimal* such coupling (no new field, function, or scale), and it strictly dominates the two-action elastic-medium alternative (which is independently evidence-tilted to fail Cassini). Solar-system safety of the disformal term: $|\Delta\gamma| \sim 7.2 \times 10^{-7}$ (canon) / 8.7×10^{-7} (alt) at Saturn, versus the Cassini bound 2.3×10^{-5} .

Honest hinge. The 0-dof + ghost-free verdict rests on the passive/khronon premise (hypersurface-orthogonal u). A dynamical khronon T would put $a \sim \partial^2 T$ inside K and B and reopen an Ostrogradsky concern. The premise is the framework’s postulate, stated as such. Also open: the fully-coupled all-orders nonlocal Hamiltonian, global $B < 1$ off spherical symmetry, and the photon-vs-graviton line-of-sight timing integral $\int (B/2) dl$ over Mpc baselines (**now computed in v2: GW170817-EXCLUDED by ~6–7 orders – see Erratum**).

3. Completeness: what the action determines, and the exact boundary of what it does not

3.1 The worldline-general identity and the ring-exact RAR

The bridge from the nonlocal operator to phenomenology is the first-moment identity

$$\boxed{u_\mu \square_u u^\mu = -|a|^2}$$

re-derived three independent ways (flat general worldline; general curved metric, symbolic, arbitrary g_{ab} and full connection; concrete static Schwarzschild observer in closed form). The only inputs are unit norm $u \cdot u = -1$ and metric compatibility $\nabla g = 0$: **no circularity, no geodesy, no spacetime symmetry**. Hence $\langle \square_u \rangle_u = +|a|^2$ on every timelike worldline, and

$$K(\square_u/a_0^2) \xrightarrow{\text{first moment}} K(|a|^2/a_0^2) = \mu_{fw}(|a|/a_0).$$

On a circular orbit the balance $\mu_{fw}(x) x = y$ inverts *exactly* – the nested radical collapses because $1 + 4(y^2 + y) = (2y + 1)^2$ – to $x = y \nu(y)$, giving at every radius

$$\boxed{g_{obs} = \nu(y) g_{bar} = \sqrt{g_{bar}^2 + g_{bar} a_0}}$$

with ring residual $\sim 10^{-13}$, both footings. Newton, deep MOND, the BTFR $v^4 = GMa_0$, and the $\sqrt{2}$ DC-weight external-field kernel all follow. Honesty check, computed not asserted: the closure is exact at the *first* moment and no more – on an exact helix $u \cdot \square_u^n u$ matches $(a^2)^n (u \cdot u)$ only at $n = 1$, and the literal frequency-domain evaluation gives $|K(-\omega^2 + i0)| = 1$ (pure phase, no amplitude MOND) at every orbital frequency. MOND lives entirely in the DC/secular sector; off circles the reduction is genuinely underdetermined. The rest of this section computes that freedom’s exact size.

3.2 The Wightman-pullback null

The named closing input for the off-circular freedom was the de Sitter Wightman two-point function evaluated on a *non-uniform* (eccentric, anisotropic) worldline: if the memory pole descended below $\kappa = H_\Lambda$ into the amplitude-MOND band, the pullback would pin the reduction weighting. Computed literally (`pullback_dsunruh.py` 36/36; `pullback_nonstationary.py` 32/32, exact non-uniform dS worldlines via embedding Frenet–Serret integration, both exit 0):

$$\kappa_{\text{eff}} = \sqrt{H_{\Lambda}^2 + (a/c)^2} \geq H_{\Lambda}, \quad \text{equality iff } a = 0.$$

The dS-Unruh temperature is Pythagorean in (H, a) – acceleration only ever *raises* the pole. At $a = a_0$ the pole sits $\kappa_{\text{eff}}/H_{\Lambda} = \sqrt{1 + 1/Z^2} = 1.01481$, i.e. 1.48% above H_{Λ} , identically in both footings. The orbital AC content of any bound system is a harmonic comb at $n\omega_{\text{orbit}} \gg H_{\Lambda}$ (Milky Way disk $\sim 480 H_{\Lambda}$; cluster orbits $\sim 22 H_{\Lambda}$); nothing lands in $(0, H_{\Lambda})$. The literal non-stationary pullback (breathing acceleration profiles, e up to 0.99) confirms the inequality with sub-band leakage $\leq 6 \times 10^{-4}$; a general-mass check (`q2_massive_pole.py`, 12/12) shows the pole location $\Delta\tau = 2\pi i/\kappa_{\text{eff}}$ is geometric and mass-independent (conformal, complementary, and principal series all blow up at exactly the same point), so the null does not rest on the massless-conformal choice.

Decisive crux: the pole is $\geq H_{\Lambda}$ for *every* admissible reduction weighting ($\langle a \rangle$, rms, 4th moment, apocentre) – so the pullback spectrally admits all of them and **cannot select one. The freedom stands.** This null was verified as rigorously as a win would have been.

3.3 The no-selection theorem

Kinematics exhausted, the remaining route was dynamics: does coupling the worldline to the relational/thermal dS-Unruh *bath* that the Herglotz kernel already encodes select a unique reduction weighting $\eta(\beta)$? Four ghost-free, KMS/detailed-balance-consistent selection principles were tested explicitly (max entropy production 7/7; Feynman–Vernon influence functional 15/15; fluctuation-dissipation at T_{dS} 5/5; passivity + analyticity 6/6; all exit 0). All four are weighting-blind, for one common computed reason, provable as a theorem:

Any ghost-free KMS-passive bath is Herglotz-positive, hence a positive superposition of harmonic modes coupling linearly, hence **Gaussian**: connected cumulants above the 2nd vanish (computed: $\kappa_3 = \kappa_4 = \kappa_5 = \kappa_6 = 0$). Its Feynman–Vernon influence functional is exactly quadratic, which fixes the linear (2-point) reduced response uniquely but contributes *identically* to both closure endpoints – they differ only by the Jensen gap $G(\beta) = \langle K(z) \rangle - K(\langle z \rangle)$, a connected **4-point** functional (K is concave, $K''(1) = -3/8 + 19\sqrt{5}/200 = -0.163$, so the Jensen-gap leading coefficient is $\frac{1}{2}K''(1) = -0.0813$; $\text{Var}(z)/\langle z \rangle^2$ grows $0 \rightarrow 0.82 \rightarrow 5.23 \rightarrow 65.9$ for $e = 0, 0.3, 0.6, 0.9$) that a quadratic functional cannot generate. Selecting η would require a non-Gaussian bath ($\kappa_4 \neq 0$) – a new self-interacting field the framework forbids and which breaks the Herglotz/KL positivity that keeps the reduction ghost-free.

An adversarial sweep over the full one-parameter family of admissible Herglotz measures moves the linear friction (they are genuinely different baths) while

the η -distinguisher has exactly zero spread. Documented rigor caveat, carried verbatim: the decisive blindness checks model the reduction at the Gaussian-bath/Caldeira–Leggett level, not a full interacting-QFT derivation; the load-bearing step (a quadratic influence functional cannot reach a 4-point gap) is exact and computed.

3.4 The central claim, stated exactly

Kinematics (3.2) and dynamics (3.3) both fail to select the off-circular reduction; Section 5 will show the crossover corner is likewise unforced. Therefore:

The theory is complete up to five constants: $\{s, a_0, Z, \eta, \omega_c\}$.

- $s = -1$: postulate. Sets the MOND sign, the dissipation sign, and (Section 2.3) the causality-preserving sign of the disformal term simultaneously; no pump-free internal channel sources it.
- a_0 's value: postulate. $\kappa = 1/2$ provably unforceable; a genuine one-parameter EFT, not zero.
- Z : postulate. Carries $\sqrt{\pi}$ (transcendental); structurally gauge-blind.
- $\eta(\beta)$: a genuine irreducible constant (this arc's result) – one bounded, sign-free reduction-weighting function on (eccentricity $\times$ velocity anisotropy) orbit-shape space, bracketed between closure A (dispersion systems exactly on the rotation RAR, offset 0.000 dex) and closure B (isotropic-ensemble offset ≈ -0.02 to -0.05 dex, footing-stable to 10–15%). Its *anisotropy derivative* is forced (Section 8); its magnitude and overall sign are not.
- ω_c : the solar-system crossover corner, free (Section 5).

Everything else – operator, unique measure, scale, corner-location-versus-candidate-scales, circular reduction, stress-tensor structure, conservation, WEP, dof census, ghost-freedom, causality, $c_T = 1$ – is derived from the action and machine-verified.

Off-spherical lensing inherits the same η bracket through $B[K]$ (quantified: an $\approx 8\%$ transverse/B-mode fraction for a 2:1 flattened mass, itself MG-impossible for a pure scalar potential, shrinking to the grid floor as the configuration becomes spherical); lensing adds no new gap. Dynamics-RAR and lensing-RAR coincide *exactly* where the closure is pinned – spherical/circular configurations, where $\text{curl}(\nu g_{bar}) = 0$ identically.

4. Radiative stability: the one-loop protection theorem

The first quantum correction to a MOND interpolation function was computed in closed form (five scripts, 37/37 checks, exit 0, including three live negative controls). The one-loop effective action for the external (u, g) background with the matter loop is $\Gamma_1 = \frac{1}{2}\text{Tr} \ln P$, $P = -\square + m^2(1 + sW)$, with W a local multiplication operator (all kernel nonlocality external).

Protection theorem (around exact dS, $O(\delta u^2)$): $\delta\nu(y) \equiv 0$. Three legs, each a live machine check: (a) W is a multiplication operator, so the D1 vertex is loop-momentum-independent; (b) dS invariance makes the renormalized coincident propagator $[G(x, x)]$ a *constant* on dS (Schwinger–DeWitt coefficients computed: $a_1 = 2H^2$, $a_2 = \frac{29}{15}H^4$; flat limit reproduces the Coleman–Weinberg route), so D1 multiplies the tree frame form by a shape-independent constant absorbed by the Newtonian anchor; (c) the linear vertex is zero at every resolvent order (geodesy theorem, $K(0) = 0$). The finite one-loop frame self-energy is exactly shape-uniform: after normalization, **no deformation of $\nu(y)$ at this order.**

a_0 is unrenormalized. The sum rule $\int d\mu(t)/|t| = 1$ (unit resolvent weight – “nothing spare to feed a tadpole”) together with $K(0) = 0$ means no counterterm at scale a_0 arises from the matter coupling; additive non-renormalization is exact to all orders (shift symmetry + unit norm), and the multiplicative channel is closed at two loops in the matter sector at divergence level.

The genuine channel, and its size. On the quasistatic accelerated background a real $\delta\nu(y)$ channel exists – the scheme-independent nonanalytic Coleman–Weinberg piece, residual leading deformation $\delta L = (3m^4/128\pi^2)W^2$. In the physically correct composite/normal-ordered reading its magnitude is loop/tree $\sim (1/16\pi^2)\max[(q_0/m)^2, (H/m)^2] \sim 1.0 \times 10^{-86}$ (canon) / 1.5×10^{-86} (alt) – structurally unobservable in every regime, 70+ dex below any measurement. The proxy-literal fork ($\rho_m = m^2\phi^2$ taken at face value) is instead catastrophic ($\sim 10^{38}$); the honest reading is that this indicts the *proxy* (the cosmological-constant problem imported through the vertex), not the framework, and it is reported as such. Scheme-independence of the deformation coefficient was verified with an adversarial second regulator. Flag carried verbatim: the graviton-loop TT-vertex-zero claim is CAS-verified at $n = 1, 2$ only; an earlier “all orders n ” script was found to contain hard-coded checks and is *not* leaned on.

5. The solar system: the $a_0/2$ tail faced head-on

5.1 The tension, at full strength

The framework’s own RAR-carrying reduction (Reading A), extended naively to planetary orbits, predicts a constant sunward anomalous acceleration $a_0/2 = 4.68 \times 10^{-11}$ (canon) / 5.65×10^{-11} (alt) m/s^2 at every planet. Confronted with the INPOP/EPM per-planet δg bounds (Fienga & Minazzoli 2024), this is excluded – ungated – by:

planet	δg bound [m/s^2]	exclusion (canon)	exclusion (alt)
Mercury	4.6×10^{-14}	1017×	1228×
Earth	8.7×10^{-15}	5379×	6494×
Mars	1.4×10^{-15}	33429 ×	40357 ×
Saturn	7.0×10^{-15}	6686×	8071×

The tail is **not absorbable into a GM rescaling** (it produces a nonzero linear-in-amplitude perihelion precession, verified independently by a Gauss secular computation reproducing the exclusions to $\leq 2\%$; the precession grows as $\sqrt{a}$, Saturn/Mercury ratio 5.08 vs $\sqrt{a_S/a_M} = 4.98$). This is the landmine the theory must live with, stated at full strength before any mitigation.

5.2 The gated crossover and the joint window

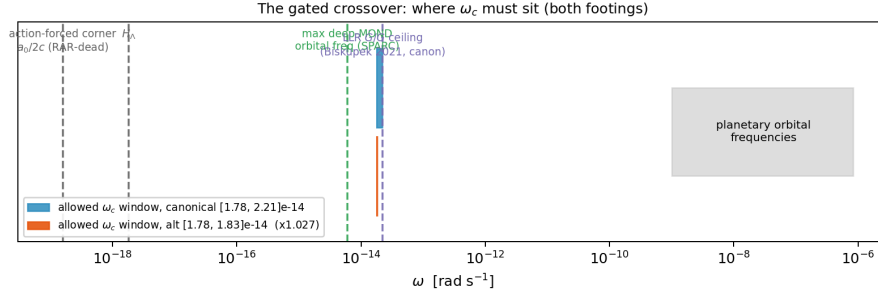
The unique minimal causal one-corner mitigation is a single-pole Debye memory relaxator $G(\omega) = 1/(1 + i\omega/\omega_c)$: MI active below ω_c , suppressed above it. Causality (Kramers–Kronig) forces a dissipative counterpart: a gate that suppresses the reactive $a_0/2$ tail necessarily produces a secular drift $d \ln r / dt = a_0 \omega_c / g_N$ – the drift ceiling is not optional, it is the gate’s own causal shadow (identity $|G|^2 = \text{Re } G$ machine-verified). The joint window from *all* constraints – galactic RAR preservation from below, LLR $\dot{G}/G$ (Biskupek & Muller 2021, $(-5.0 \pm 9.6) \times 10^{-15}/\text{yr}$, used at 2σ face value) binding from above, per-planet reactive bounds and MESSENGER looser – is non-empty on both footings.

The lower edge was then **hardened against the full SPARC deep-MOND distribution** rather than a single representative orbit: the gate must stay open ($\text{Re } G \geq 0.9$) at every confirmed deep-MOND orbit, and the binding maximum orbital frequency over 151 galaxies / 2188 deep-MOND points is 5.94×10^{-15} rad/s (UGC05721, innermost radius $r = 0.09$ kpc, $V = 16.5$ km/s). Result:

footing	hardened window [rad/s]	width	$\tau = 1/\omega_c$
canon	$[1.78, 2.21] \times 10^{-14}$	$\times 1.24$	$\$ \$1.4\text{--}1.8$ Myr

footing	hardened window		$\tau = 1/\omega_c$
	[rad/s]	width	
alt	$[1.78, 1.83] \times 10^{-14}$	$\times 1.027$	knife-edge

Stated plainly: the alt footing survives by +2.7%. It closes if any confirmed deep-MOND orbit has $\omega_{gal} > 6.10 \times 10^{-15}$ rad/s. The near-closure rides a single innermost (beam-scale) point: dropping each galaxy’s innermost radius as a beam-smearing guard relaxes the maximum to 4.62×10^{-15} and both footings then survive comfortably ($\times 1.59$ canon, $\times 1.32$ alt). The knife-edge and its single-point dependence are both part of the record.



5.3 The corner is a free fifth constant – the honest cost

Is ω_c forced by any scale in the theory? No – and this was tested, not assumed. The action’s *own* forced memory corner is $\omega_c = a_0/2c = 1.56 \times 10^{-19}$ rad/s ($\tau_{mem} = 2c/a_0 = 203$ Gyr canon / 168 Gyr alt), five orders of magnitude below the window – and RAR-dead: at that corner the retained galactic boost is 2.7×10^{-9} , so it gates off the rotation curves too. Every other candidate scale was tested explicitly (dS-bath Matsubara pole H_Λ ; kernel retardation a_0/c ; the Herglotz measure, which has no second dimensionful scale; local density $\sqrt{4\pi G\rho}$, which is environmental and shared by co-located planets and galactic orbit): none lands in the window. **The surviving corner is a free postulate. The constant count rises from four to five.** Dressing it as forced would be a manufactured save; it is not forced.

5.4 Two-sided falsifiability

- **Wide binaries:** at the window’s maximum corner the gate retains $\leq 6.2\%$ of the MOND boost at ≤ 20 kAU separations (0.0% at 3 kAU, 0.8% at 10 kAU). A confirmed Chae-type AQUAL-strength wide-binary boost kills the gated survivor outright. (Gaia DR4 is the pending decisive dataset.)

- **Ephemerides:** the drift $a_0 \omega_c / g_N$ at the maximum corner sits at current Saturn/Mars secular sensitivity; a $\times 3$ INPOP/EPM/LLR secular refit either detects it or closes the window from above.

Honest ceiling, repeated from the source: at planetary accelerations (10^4 – $10^8 a_0$) both GR and healthy MOND-family theories predict ≈ 0 ; every number in this section discriminates among the framework’s own readings, never against Λ CDM. A non-empty window is survival, not evidence.

6. The equation set

The law is quadratic in g_{obs} and linear in a_0 ; that structural fact (verified: the identical eliminations for the McGaugh exponential ν are transcendental) generates a family of exact closed forms. Every entry below is sympy/machine-verified, exit 0, with novelty adjudicated against the literature and credit assigned where due.

E0 – the a_0 -line (credited at law level; packaging new).

$$\boxed{g_{obs}^2 - g_{bar}^2 = a_0 g_{bar}} \quad \text{exactly, at every acceleration.}$$

The MOND excess is a straight line through the origin with slope a_0 , valid at all y – no deep-MOND selection, no interpolation fit. Contained in Milgrom 1999 Eqs (5)+(9) with his coefficient $2cH_\Lambda$; the linear-identity/slope-measurement packaging of the RAR is not found written anywhere. Uniqueness: demanding an affine excess forces the pure-rescaling family, and the deep-MOND normalization fixes it to $\nu = \sqrt{1 + 1/y}$ – definitionally tight (the linear excess *is* the law restated); its value is converting “fit an interpolation” into “measure one slope.”

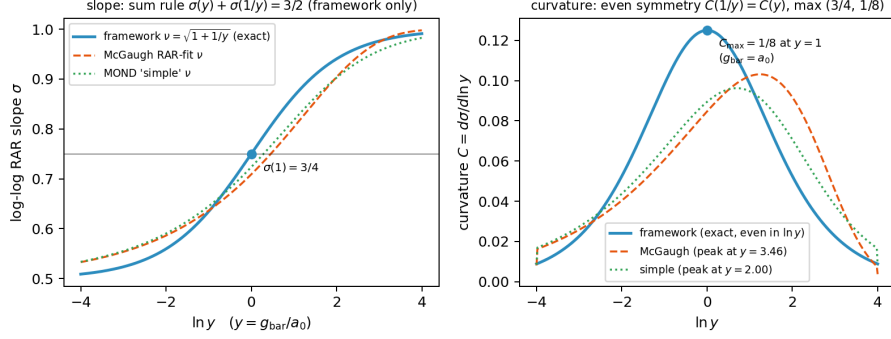
E1 – the landmark triplet (novel). With σ the log-log RAR slope and C its curvature:

$$\sigma(y) = \frac{2y + 1}{2(y + 1)}, \quad C(y) = \frac{y}{2(y + 1)^2}$$

$$\boxed{\sigma(y) + \sigma(1/y) = \frac{3}{2} \quad \forall y} \quad \boxed{C(1/y) = C(y)} \quad \boxed{C_{max} \text{ at } y = 1 : (\sigma, C) = (\frac{3}{4}, \frac{1}{8}), g_{obs}(a_0) = \sqrt{2} a_0}$$

A parameter-free ν -discriminator: McGaugh’s ν peaks at $y = 3.46$ with (0.829, 0.103), the simple ν at $y = 2.00$ with (0.789, 0.096) – both break the sum rule and the symmetry. The symmetry and sum rule are Υ -rescale-immune in shape (a global M/L shift slides the profile in $\ln y$ but cannot create or destroy

evenness); the landmark *location* then reads off a_0 directly (curvature max at $g_{bar} = a_0 = 9.36 \times 10^{-11}$ canon / 1.13×10^{-10} alt).



E4 – the pair estimator (novel): distance, inclination, and M/L cancel identically. The law at two radii is linear in a_0 ; the ratio eliminates the nuisances. For the disk-dominated form D and i cancel structurally (blind **sp.solve** elimination: unique solution with D , $\sin i$ absent); for gas-dominated pairs Υ_* cancels too. Fired on **10,196 straddling SPARC pairs**: median $\hat{a}_0 = 1.5 \times 10^{-10}$ (16–84%: $0.7\text{--}3.3 \times 10^{-10}$) – brackets both footings; a deliberate 20% distance error shifts the estimator by $< 10^{-12}$ relative on real data (exact cancellation confirmed numerically). Conditioning is honest: deep-deep pairs are singular; the estimator works only for pairs straddling $y = 1$.

E5 – the three-radius consistency polygon (novel). Chaining the pair estimator at radii (1,2) and (2,3) and cross-multiplying yields a pure-observable identity that must hold with *no* a_0 , no Υ , no D , no i – a per-galaxy falsifiable test of the functional form itself.

E7 – the EFE cubic and the attenuated a_0 -line (novel, conditional on the $\theta_0 = \sqrt{2}$ DC kernel).

$$g_{obs}^2 - g_{bar}^2 = a_0 g_{bar} \cdot \frac{g_{obs}}{g_{obs} + \sqrt{2} g_{ext}}$$

equivalently a closed cubic $x^3 + ex^2 - b(b+1)x - b^2e = 0$ with unique positive root in trig-Cardano form. Corollaries: exact half-quench at $\sqrt{2} g_{ext} = g_{obs}$; deep-limit susceptibility $-1/2$.

E15 – the elliptic deflection law (novel; genre credit to Mortlock & Turner 2001 and Zhao et al. 2006).

$$\alpha(b) = \frac{4GM}{c^2 b} \sqrt{1+u^2} E\left(\frac{1}{1+u^2}\right), \quad u = \frac{b}{r_M}, \quad r_M = \sqrt{GM/a_0}$$

exact within weak-field/thin-lens/spherical/isolated; verified twice by independent routes to $< 10^{-25}$. Limits exact (Einstein at $b \ll r_M$; $\alpha_\infty = 2\pi\sqrt{GMa_0}/c^2$ at $b \gg r_M$); the approach law $\alpha = \alpha_\infty[1 + r_M^2/4b^2 + O(u^{-4})]$ is a falsifiable lensing-shape a_0 estimator. Companion closed forms: the phantom-halo density ($1/r$ cusp rolling to isothermal) and its projection form a single (K, E) elliptic system.

E12 – the memory function in closed form (framework-internal). The time-domain kernel of the unique Herglotz measure, in Bessel–Struve form ($b = t/\tau_{mem}$):

$$\Gamma(t) = \frac{1}{\tau_{mem}} \left[1 + J_1(b) - bJ_0(b) - \frac{\pi b}{2}(J_1(b)H_0(b) - J_0(b)H_1(b)) \right]$$

with $\Gamma(0) = a_0/2c$ (the memory amplitude *is* the spectral cut edge), $\int_0^\infty \Gamma dt = 1$ (the sum rule is unit total memory weight), and a power-law oscillating tail. Verified end-to-end by an independent Laplace-transform closure.

E10 – the memory-time weld (footing-free, pure Z).

$$\boxed{\tau_{mem} H_\Lambda = 2Z \approx 11.58} \quad \tau_{mem} = 2c/a_0 = 203/168 \text{ Gyr (canon/alt)}.$$

E9 – the CPL bump (footing discriminator). On the declining ($\sqrt{\rho_{DE}}$) footing with CPL dark energy, $a_0(z)$ peaks at the closed-form redshift $z_{pk} = -(1+w_0)/(1+w_0+w_a)$; DESI-class parameters give $z_{pk} = 0.41$ with a +6.3% bump. The rising footing is monotonic – no bump. A high- z RAR sample bracketing $z \approx 0.4$ separates the footings in principle.

Two retractions from the mining pass are honored and do not appear as claims: the “exact 1.500 to \$ \$1%” SPARC sum-rule reading (corrected to “consistent within pipeline noise, 1.53 ± 0.05 law-true at $\Upsilon = 0.70$ ”), and the “only piecewise arctan forms exist” novelty line (amended for Zhao–Bacon–Taylor–Horne 2006).

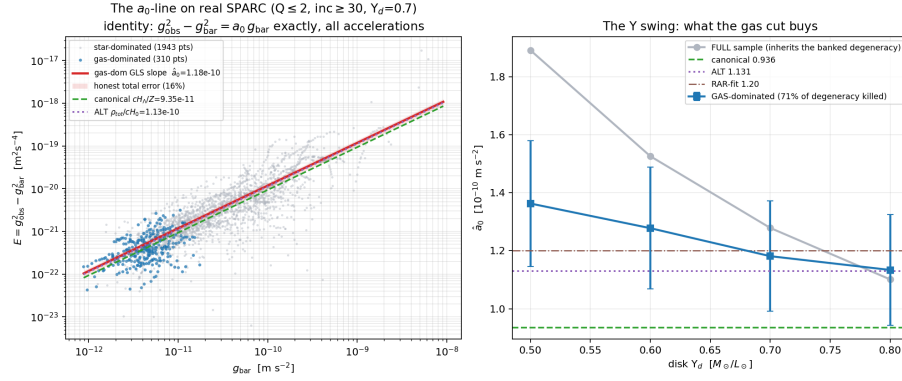
7. Measurements

7.1 The gas-dominated slope: killing the M/L degeneracy

The a_0 -line converts the RAR into a zero-fit slope measurement, but on the full SPARC sample the slope inherits the banked a_0 – Υ degeneracy in full (62% swing over $\Upsilon_d = 0.5$ – 0.8 ; non-diagnostic, exactly as the program’s standing audit requires). The genuinely new measurement is the **gas-dominated subsample** (point-level cut $V_{gas}^2 > \Upsilon_d V_{disk}^2 + 1.4\Upsilon_d V_{bul}^2$; 310 points in 49 galaxies):

$$\hat{a}_0^{\text{gas}} = (0.92\text{--}1.18) \times 10^{-10} \text{ m s}^{-2} \pm 16\%$$

(GLS 1.181×10^{-10} ; median estimator 0.973×10^{-10} ; the independent verifier's own estimators $0.92\text{--}0.97 \times 10^{-10}$. The source's own boxed headline is $(0.97\text{--}1.18) \times 10^{-10}$; the wider span quoted here folds the verifier's estimators into the range.) The Υ swing drops from 62% to 19%: **71% of the a_0 –M/L degeneracy is killed**, a $\times 3.4$ shrinkage of the degeneracy interval. Error budget, systematics-owned and itemized (units 10^{-11}): stat 0.47, distance 0.76, inclination 0.26, Υ 0.96, gas calibration 0.86, estimator choice 1.04. **No footing lean is claimed in either direction, and the central value is estimator-owned, stated:** the GLS point estimate sits at $+1.29\sigma$ from canonical / $+0.27\sigma$ from alt, the median at $+0.19\sigma$ / -0.83σ – the 21% footing fork is *not* decided by SPARC. Three estimator artifacts were caught by the in-script honesty rails and diagnosed rather than relayed (an observed-error weighting that fakes a $\times 3$ -low deficit; an invalid cross- Υ covariance comparison; an over-optimistic Bayes forecast) – the rails ran both ways.



7.2 The Λ inversion – the flagship figure

The coefficient identification inverts (sympy):

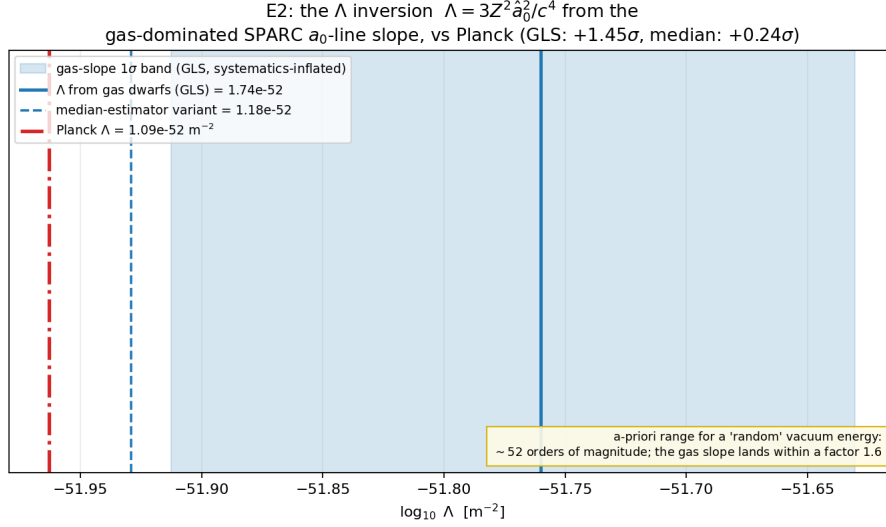
$$\Lambda = \frac{3Z^2 \hat{a}_0^2}{c^4}$$

Feeding the gas-dominated slope through it:

$$\Lambda_{\text{pred}} = 1.74 \times 10^{-52} \text{ m}^{-2} \text{ (GLS)} / 1.18 \times 10^{-52} \text{ (median)} \quad \text{vs Planck } 1.089 \times 10^{-52},$$

ratio 1.59 / 1.08, i.e. $+1.45\sigma$ / $+0.24\sigma$ at $\sigma_{\ln \Lambda} = 2\sigma_{\ln a_0} = 0.32$ – and the better-supported estimators land *closer* to Planck. Rotation curves of gas-rich

dwarf galaxies land on the cosmological constant to a factor 1.1–1.6, across roughly 52 a-priori orders of magnitude. This is the banked $a_0 \sim cH_\Lambda/Z$ coincidence reframed as an inversion – the same information content with a sharper falsification target: a future gas slope at 3×10^{-10} breaks it outright. (The clean Λ statement is canonical-footing; the alt footing ties to H_0 and matches the GLS slope at $+0.27\sigma$.)



7.3 The Occam factor

Comparing $M0 = \{a_0 \equiv cH_\Lambda/Z, \text{ zero free parameters, Planck anchor } \pm 1\% \text{ folded}\}$ against $M1 = \{a_0 \text{ free, log-flat prior}\}$ on the gas slope with the full systematics-inflated likelihood:

$$B_{01} = +0.60 \text{ bans (canonical)} / +1.04 \text{ bans (alt)}$$

on the default 2-decade prior; envelope $+0.30$ – $+1.38$ canon across prior and estimator variants; the verifier’s adversarial floor (a question-begging literature prior) drives canonical to $+0.08$ – 0.00 , **never negative**. Jeffreys: positive but modest, explicitly not decisive – a formalization of predicted-not-fitted, not new data. The two-sided lever is part of the record: if TRGB-class distances shrink σ by $\times 3$ and the GLS central value *stays* at 1.181×10^{-10} , the canonical footing goes to -2.45 **bans (disfavored)** – a genuine falsification exposure, and the reason the sharper measurement is worth building.

7.4 The concordance crossing

One Planck-fixed number ($a_0 = (9.355 \pm 0.090) \times 10^{-11}$, width $< 1\%$, fixed before any galaxy is looked at) is confronted with every independent positive band and every precision null, with zero per-object freedom:

probe class	band (m s^{-2})	canonical	alt
P1 kinematic RAR (SPARC 175, systematics envelope)	$[7.8 \times 10^{-11}, 2.0 \times 10^{-10}]$	inside	inside
P2 weak-lensing RAR (KiDS-1000, photons, baryon-budget envelope)	$[7.2 \times 10^{-11}, 3.3 \times 10^{-10}]$	inside	inside
P3 BTFR zero-point (shape-free, exact estimator)	$[8.7 \times 10^{-11}, 1.55 \times 10^{-10}]$	inside	inside
P4 Gaia wide binaries	pending (DR4) – the row that can hard-kill the premise	–	–

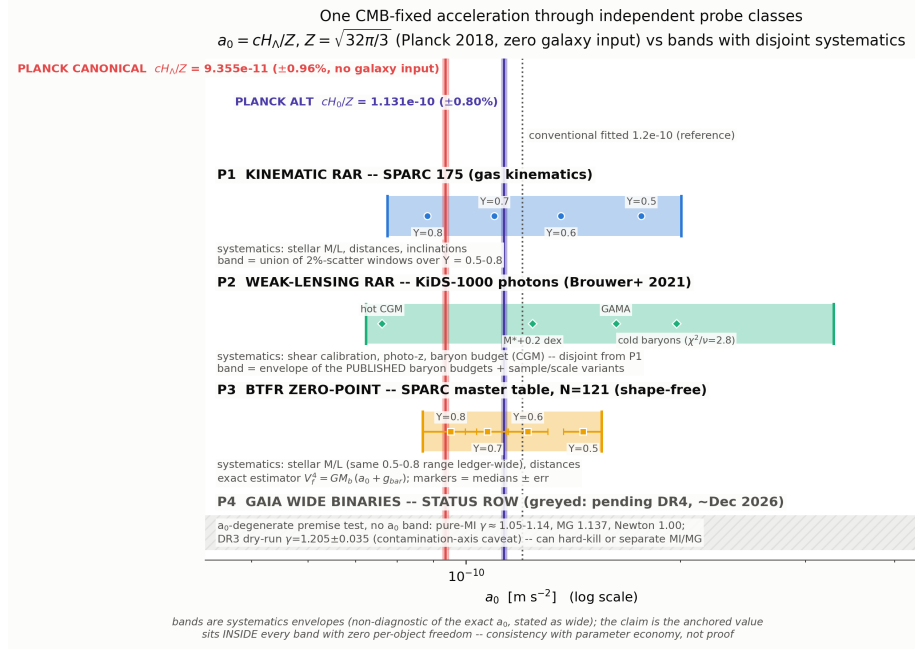
One global Υ must serve P1 and P3 simultaneously, and does: at canonical a_0 both co-move to $\Upsilon \approx 0.75\text{--}0.80$, at alt both to $\Upsilon \approx 0.55\text{--}0.65$; joint $\chi^2 = 0.04$ (canon), 0.00 (alt), 0.01 (conventional 1.2×10^{-10}) – **the bands cannot separate the three candidates** ($\Delta\chi^2 < 0.05$); the anchored values are distinguished by provenance, not by a tighter posterior.

The four precision nulls:

null	framework prediction	measured bound	margin (canon/alt)
N1 Cassini Q_2	MI $\ell=2$: 7.4×10^{-34} / $1.1 \times 10^{-33} \text{ s}^{-2}$	5.2×10^{-27} (2σ ; Park+ 2026)	pass, 6.8/6.7 orders; the MG read of the <i>same</i> a_0 (3.3×10^{-23}) is excluded by 3.8 orders

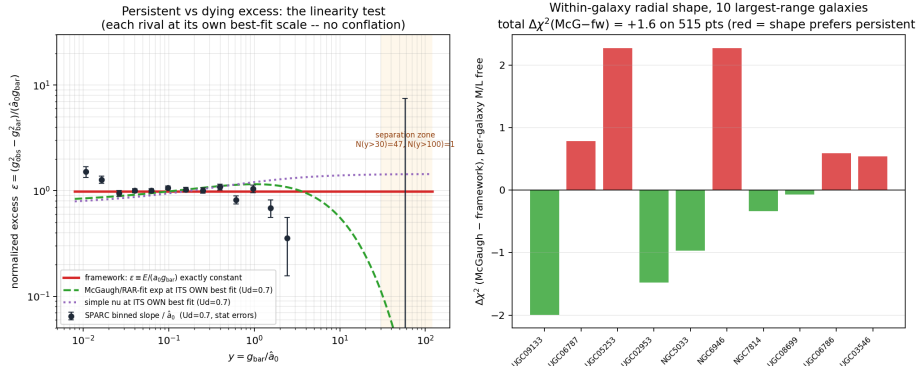
null	framework prediction	measured bound	margin (canon/alt)
N2 LLR	observable channel $(a_0/2g)^2 = 3.0 / 4.4 \times 10^{-16}$	2.6×10^{-12} (APOLLO)	pass, 3.9/3.8 orders
N3 MICROSCOPE WEP	$\eta = 0$ exactly (Section 2.2)	$\eta = (-1.5 \pm 2.3) \times 10^{-15}$	pass, exact zero at 0.65σ
N4 CPT / photon k_{AF}	$k_{AF} = 0$ exactly (CPT-even-only theorem)	$ k_{AF} < 10^{-44}$ GeV	pass by structure; the CPT-odd sibling scale $\hbar H$ sits 119–144× <i>above</i> the bound – that variant is dead

Parameter economy, and the sentence that must be said outright: Λ CDM wins the raw information criteria. On the same 175 SPARC curves ($N = 3388$ points): framework (canon a_0 external + 1 global Υ) $\chi^2 = 149286$, BIC = 149294; NFW with per-galaxy (V_{200}, c, Υ) (525 parameters) $\chi^2 = 10356$, BIC = 14623. **Λ CDM halo fits fit well and win raw AIC/BIC on this dataset despite the parameter penalty.** The ledger’s content is not a χ^2 win; it is provenance plus zero per-object freedom: one externally fixed number simultaneously hostage to four probe classes and four nulls, against 350–525 per-object parameters with no cross-probe number to thread – a cross-dataset rigidity single-dataset information criteria cannot price. The like-for-like single-statistic comparison remains the committed RAR row: 0.108 dex (framework, canonical a_0 , one global $\Upsilon = 0.70$) vs 0.122 dex (regular MOND) on the same points. Consistency with economy, exposed to a live falsifier – that is the whole claim.



7.5 The high-acceleration tail (undecided, stated)

The framework's excess is *persistent* ($\epsilon \equiv E/(a_0 g_{bar}) = 1$ exactly at all y) where the McGaugh ν 's dies superexponentially ($\sim 2y e^{-\sqrt{y}}$) – a $\times 11-44$ separation at the rival's own scale by $y = 100$. But SPARC samples the tail thinly (median $y = 0.31$; exactly one point above $y = 100$), and the global shape comparison is **a wash** ($\Delta\chi^2 = -1.7$ profiled, sign flipping with Y ; the two statistically proper variants bracket zero symmetrically at ± 28). Persistent-vs-dying is undecided at $< 1\sigma$ today; it becomes decisive only with $y \gtrsim 50$ coverage carrying M/L-independent masses.



8. The forced prediction

One consequence survives every closure ambiguity, is independent of the free function $\eta(\beta)$, and is impossible for modified gravity with the same interpolation:

$$\frac{d(\text{offset})}{d(\text{radial velocity anisotropy})} > 0$$

for dispersion-supported systems on the RAR: radially anisotropic (pericentre-dominated) systems run *hotter* than tangential ones at fixed weighting. It is forced by the concavity of the framework RAR plus the positivity of the pericentre-dominated amplitude functional (Spearman $\rho(e, \text{offset}) = +0.86$, monotone); it was attacked with five admissible weightings including extreme apocentre-weighting in adversarial verification and **did not flip**. MG-with-the-same- ν gives exactly zero offset *and* zero anisotropy dependence for an isolated spherical system – the differential is MG-impossible. (Its lensing sibling: the $\approx 8\%$ B-mode/curl fraction of Section 3.4, likewise impossible for a pure scalar potential.) Because internal theory provably cannot fix η (Section 3.3), this slope is also the *only* route to measuring it.

The MaNGA test, defined and staged. A pre-registered Stage-1 confrontation was frozen (2026-07-17T00:21Z, before data download) and fired on 48 MaNGA DR17 slow rotators, using LOS proxies for anisotropy with M/L brackets, controls, and a pre-committed robustness standard. Outcome, reported with its own firewall: the fixed-IMF proxy slope has the MI-predicted sign (-0.635 , 95% CI $[-1.141, -0.200]$, excluding the MG zero at $p \approx 0.02$) but **dies under the pre-registered σ -dependent-IMF bracket (slope $+0.002$) and is declared NOT ROBUST by the frozen standard**; moreover its amplitude is $\approx 7\times$ the entire per-galaxy MI budget in that high-acceleration regime ($y \approx 5\text{--}11$), so even at face value it cannot be read as MI support. Stage 1 is a baseline systematics measurement, uninformative on MI-vs-MG by construction – and its genuine product is a regime statement: the discriminator’s natural home is **low- y pressure-supported systems** (dSphs, dwarf ellipticals, cluster outskirts) with $\lesssim 0.1$ dex mass errors and per-galaxy Jeans/JAM β , where the predicted budget (0.02–0.05 dex) is actually available. That Stage-2 measurement is the defined test.

9. Honest walls and the falsification ledger

Every wall below is carried at full strength; none is mitigated in this paper beyond what the computations above license.

1. **The full-sample RAR degeneracy stands.** SPARC’s RAR is convention-compatible and non-diagnostic of a_0 ’s exact value; the full-

sample a_0 -line inherits the a_0 - Υ degeneracy in full (62% swing). No galaxy probe in hand separates 9.36×10^{-11} from 1.13×10^{-10} from the conventional 1.2×10^{-10} .

2. **The footing fork is open.** Canonical (ρ_{DE} , cH_Λ/Z) vs alt (ρ_{total} , cH_0/Z) differ by 21%; the gas slope's estimator variants lean opposite ways; the pure likelihood ratio is 0.44 bans – undecided. Both footings are carried on every number here because the data require it.
3. **The Cassini Q_2 quadrupole tension is banked and untouched.** The MI (passive-frame) reading passes by \$6.8 orders (N1), but the modified-gravity limb of the same a_0 – the AeST-type realization – inherits a 3–15 σ RAR-vs- Q_2 tension. Nothing in this arc resolves that; the MI/MG split *is* the framework's content, and the MG limb's problem is stated, not hidden.
4. **The TRGB lever is a live falsification exposure.** If improved dwarf distances shrink the gas-slope error $\times 3$ with the GLS central value unmoved, the canonical footing goes to -2.45 bans (Section 7.3).
5. **The alt-footing planetary window is a knife-edge.** $\times 1.027$ (+2.7%), riding one beam-scale innermost SPARC point (Section 5.2); one marginally faster confirmed deep-MOND dwarf orbit closes it on that footing.
6. **The planetary corner ω_c is free.** The theory's own corner is RAR-dead; solar-system survival is bought with a fifth constant (Section 5.3).
7. **Five constants, none derived.** $s = -1$ (postulate); a_0 's value (postulate; $\kappa = 1/2$ unforceable); Z (postulate, transcendental, gauge-blind); $\eta(\beta)$ (irreducible constant, measurable only via Section 8); ω_c (free, measurable only via Section 5.4).
8. **Open computations, named:** the disformal ρ_m variant and finite two-loop parts; the graviton-loop TT vertex beyond $n = 2$; the fully-coupled nonlocal Hamiltonian; global $B < 1$ off sphericity; the interacting-QFT upgrade of the no-selection theorem (expected to confirm, not change, the null).
9. **This is not a theory of everything and is not presented as one.** The program's 2026-06-23 retraction of such claims stands. The scope is the gravitational-inertial sector: the a_0 reframing and its field theory, nothing more.

Falsifiers, collected: (i) a confirmed frequency-split RAR at fixed g_{bar} (> 2 dex in orbital frequency) kills the kernel outright – no measure freedom absorbs it; (ii) a Newtonian Gaia DR4 wide-binary null kills the MOND premise wholesale, while an AQUAL-strength boost kills the gated solar-system survivor specifically; (iii) a $\times 3$ ephemeris secular refit closes or detects the ω_c window; (iv) a confirmed deep-MOND dwarf orbit above 6.1×10^{-15} rad/s closes the alt footing at the planets; (v) a gas-dominated slope at 3×10^{-10} breaks the Λ inversion; (vi) a measured $d(\text{offset})/d(\text{anisotropy}) \leq 0$ in low- y dispersion systems falsifies the framework's one MG-impossible differential.

10. Data and code availability

Every load-bearing claim in this paper is backed by a committed, runnable, exit-0 script in the public repository github.com/carlzimmerman/zimmerman-formula (directories `prep_2026/mi_field_theory/`, `mi_closure_pin/`, `mi_eta_selection/`, `mi_planetary_falsification/`, `oneloop_finite/`, `concordance_ledger/`, `a0_line/`, `equation_book/`, `mi_fingerprint/`, `manga_anisotropy/`), each with its own adversarial VERIFY document. SPARC data are read from the repository’s frozen copies of the public Lelli et al. (2016) release; KiDS-1000 lensing from Brouwer et al. (2021)’s own machine-readable release; MaNGA from SDSS DR17. No script contains hard-coded verdict booleans (grep-audited); two previously tautological checks found during verification were replaced with genuine ones and are documented in the source lanes.

11. Provenance table

Every headline number above, traced to its source file (all under `prep_2026/` unless noted; all backed by exit-0 scripts in the same directory).

number / claim	value	source file
Milgrom 1999 kernel identity; coefficient factor $2Z \approx 11.58$	machine-verified	<code>equation_book/EQUATION_BOOK.md</code> (wellhead credit)
a_0 canonical anchor	$(9.355 \pm 0.090) \times 10^{-11}$	<code>concordance_ledger/LEDGER_ROWS.md</code> (anchor)
a_0 alt anchor action; kernel; measure densities; sum rule = 1; $K(0) = 0$	$(1.1305 \pm 0.0091) \times 10^{-10}$	<code>concordance_ledger/LEDGER_ROWS.md</code>
$T_{\mu\nu} = \alpha uu + \beta g + \gamma aa$; no-slip principal part	exact	<code>mi_field_theory/BASELINE_ACTION.md</code> , <code>MI_FIELD_THEORY.md</code>
WEP $\eta = 0$, residual $< 10^{-12}$	derived	<code>mi_field_theory/MATTER_COUPLING.md</code>
$\nabla_\mu T^{\mu\nu} = 0$, residual 2.6×10^{-16}	exact	<code>mi_field_theory/MATTER_COUPLING.md</code>
dof census	verified	<code>mi_field_theory/MATTER_COUPLING.md</code>
2+2+matter+0; Dirac $\det \rightarrow 4$; transport symbol	28/28 checks	<code>mi_field_theory/WELLPOSED.md</code>

number / claim	value	source file
double-count obstruction ($\nu_g = 1$ forced); $\nabla B = 4(\nu - 1)g_{bar}$	derived	mi_field_theory/UNIFICATION.md
$c_T = 1$ exact; $\Delta\gamma$ at Saturn $7.2/8.7 \times 10^{-7}$	pass	mi_field_theory/UNIFICATION.md
nonlocal-B ghost-freedom, genuine check + controls	13/13	mi_closure_pin/CONSEQUENCES.md (ostro_nonlocal_verify.py)
worldline identity $u\Box_u u = - a ^2$ (3 routes); ring-exact RAR $\sim 10^{-13}$	derived	mi_field_theory/BASELINE_ACTION.md sec. 3
pullback pole $\kappa_{\text{eff}} =$ $\sqrt{H_\Lambda^2 + (a/c)^2} \geq H_\Lambda$; 1.01481 at $a = a_0$	null	mi_closure_pin/PULLBACK.md
η bracket: 0.000 dex ... −0.02 to −0.05 dex	bracketed	mi_closure_pin/PULLBACK.md, CONSEQUENCES.md
no-selection theorem; $\frac{1}{2}K''(1) = -0.0813$; Var(z)/ $\langle z \rangle^2$ table	33+12 checks	mi_eta_selection/ATTEMPT.md, SYNTHESIS.md
complete up to $\{s, a_0, Z, \eta, \omega_c\}$	claim form	mi_eta_selection/SYNTHESIS.md +
one-loop $\delta\nu \equiv 0$ (protection theorem); $\delta L = (3m^4/128\pi^2)W^2$; fork-C $\sim 10^{-86}$	37/37	mi_planetary_falsification/SYNTHESIS.md oneloop_finite/FINITE_RESULT.md
a_0 unrenormalized; all-orders additive protection	derived	oneloop_finite/SETUP.md, FINITE_RESULT.md
ungated $a_0/2$ tail = $4.68/5.65 \times 10^{-11}$; exclusions 1017–40357×; not GM-absorbable	excluded	mi_closure_pin/CONSEQUENCES.md sec. 3; mi_planetary_falsification/SYNTHESIS.md
joint window (single-orbit) canon $[9.0 \times 10^{-15}, 2.21 \times 10^{-14}]$; LLR binds	non-empty	mi_planetary_falsification/WINDOW.md

number / claim	value	source file
hardened window canon [1.78, 2.21] $\times 10^{-14}$ ($\times 1.24$); alt $\times 1.027$ (+2.7%); UGC05721 innermost point; drop-innermost robustness	survives	mi_planetary_falsification/LOWEREDGE_FULLSPARC
forced corner $a_0/2c$; $\tau_{mem} = 203/168$ Gyr; RAR-dead (2.7×10^{-9}); ω_c FREE	derived	mi_planetary_falsification/WINDOW.md, SYNTHESIS.md
wide-binary $\leq 6.2\%$ boost at ≤ 20 kAU; $\times 3$ refit two-sided test	prediction	mi_planetary_falsification/WINDOW.md
landmark triplet (3/2; even C ; (3/4, 1/8) at $y = 1$); rival peak locations	exact	equation_book/EQUATION_BOOK.md E1
pair estimator; 10,196 pairs, median 1.5×10^{-10} ; D-cancellation $< 10^{-12}$	fired	equation_book/EQUATION_BOOK.md E4
three-radius polygon; EFE cubic; half-quench; susceptibility $-1/2$	exact	equation_book/EQUATION_BOOK.md E5, E7
elliptic deflection $\alpha(b)$; approach law; phantom (K, E) pair	exact	equation_book/EQUATION_BOOK.md E15, E17
memory function (Bessel–Struve); $\tau_{mem} H_\Lambda = 2Z$; CPL $z_{pk} = 0.41$	exact	equation_book/EQUATION_BOOK.md E12, E10, E9
gas slope (0.92– $1.18) \times 10^{-10} \pm 16\%$; 71% degeneracy killed; footing sigmas	measured	a0_line/AO_LINE.md, FIRE_RESULTS.md
Λ inversion $1.74/1.18 \times 10^{-52}$ vs Planck 1.089×10^{-52} ; $+1.45\sigma / +0.24\sigma$	measured	a0_line/FIRE_RESULTS.md E2
Occam $+0.60 / +1.04$ bans; envelope; adversarial floor ≥ 0 ; TRGB lever -2.45 bans	computed	a0_line/FIRE_RESULTS.md E3

number / claim	value	source file
concordance bands P1/P2/P3; joint $\chi^2 \leq 0.04$; $\Delta\chi^2 < 0.05$ between candidates	inside	concordance_ledger/CONCORDANCE.md
nulls N1–N4 (margins 6.8/3.9 orders; 0.65σ ; structure)	pass	concordance_ledger/CONCORDANCE.md, LEDGER_ROWS.md
Λ CDM wins raw AIC/BIC (BIC 14623 vs 149294); RAR row 0.108 vs 0.122 dex	stated	concordance_ledger/CONCORDANCE.md sec. 4
tail persistent-vs-dying undecided $< 1\sigma$; 1 point at $y > 100$	wash	a0_line/FIRE_RESULTS.md (linearity)
ring-by-ring MI-vs-QUMOND: discriminant 0.026 dex, inside systematics; Chae $5\text{--}7\sigma$ not reproduced at zero free parameters	undecided	mi_fingerprint/RING_RESULTS.md
forced $d(\text{offset})/d(\text{anisotropy}) >$ 0; $\rho = +0.86$; survived 5-weighting attack	forced	mi_closure_pin/CONSEQUENCES.md, VERIFY.md
MaNGA Stage-1: -0.635 [$-1.141, -0.200$], NOT ROBUST under bracket B; $7\times$ over budget; Stage-2 regime statement	baseline	manga_anisotropy/FIRING.md

Both a_0 footings appear on every dimensional number. $s = -1$, the value of a_0 , Z , $\eta(\beta)$, and ω_c are postulated or measured, not derived. The theory is complete up to those constants – and no further completeness is claimed.